Customer Shopping Behavior Analysis

1. Project Summary

This project analyzes customer shopping behavior using a dataset of 3,900 transactional records. We aim to gain insights by identifying behavioral and spending patterns, as well as preferred categories and high-profitability products. These findings will help the business make informed decisions to drive growth.

2. Dataset Summary

- **Rows:** 3900
- **Columns:** 18 columns with the following features:
  - **Customer Demographics:** Customer ID, Age, Gender, Location, Subscription Status
  - **Purchase Details:** Item Purchased, Category, Purchase Amount (USD), Size, Color, Season, Review Rating, Shipping Type
  - **Shopping Behaviour:** Discount Applied, Promo Code Used, Previous Purchases, Payment Method, Frequency of Purchases
- **Missing/Null values:** 37 missing/null values in the review ratings column.

3. Exploratory Data Analysis(EDA)

To get a general understanding of the dataset, we used Python for data exploration.

1. **Load the Data:** Using the pandas library, we loaded the dataset into our working environment.
2. **Exploration:** To understand the structure and features of the dataset, the `head()`, `describe()`, `info()`, and `isnull()` functions were used.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payn Met
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Ve
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	(
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	C (

3. **Column Standardization:** We standardized the column names to the ***snake_case*** style to enhance readability and simplify documentation.
4. **Handling Missing Values:** The 37 missing values in the review ratings were replaced using the column's median. The median was chosen because the data was found to be relatively well-distributed and not heavily skewed or distorted.

5. **Feature Engineering:** The following features were added to enable easy analysis of the dataset in later sections:
    a. ***age_group:*** added to facilitate customer categorization by age.
    b. **Frequency_of_purchases:** added to enable numerical analysis of purchase frequency.
6. **Data Redundancy:** The '*discount_applied*' and '*promo_code_used*' features were found to be redundant (exact copies), so we dropped the '*promo_code_used*' feature.
7. **Connecting to MySQL Database:** Finally, the cleaned data was imported into a MySQL database.

4. Data Analysis Using MySQL

We found several important business insights from our analysis, which are detailed below:

1. **Revenue by Gender:** We were able to discover that males generate over double the amount of revenue females generate.

gender	revenue_generated
Male	157890
Female	75191

2. **High Spending Discount Users:** With 839 rows of data returned, these are customers that used a discount and still spent more than the average purchase amount.

customer_id	purchase_amount_usd
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90

3. **Top 5 Products by Review Ratings:** Gloves are the most reviewed product in the store.

	products	avg_review_rating
▶	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Skirt	3.78

4. **Standard vs Express Shipping Amounts:** Express shipping leads, though by a small margin, in terms of average shipping amounts.

	shipping_type	avg_purchase_amount
▶	Express	60.48
	Standard	58.46

5. **Highest Spender:** Non-subscribed customers spend more both on average and in total.

	custumer_type	avg_spend	total_revenue
▶	No Subscription	59.8651	170436
	Subscribed	59.4919	62645

6. **Top Products Bought with Discounts:** The 5 products with the highest percentage of purchases with discounts applied were identified.

	product	discount_purchase	no_discount	percent_discount_purchases
▶	Sneakers	4322	4313	50.05
	Hat	4596	4779	49.02
	Sweater	4610	4852	48.72
	Coat	4470	4805	48.19
	Par Coat	4837	5253	47.94

7. **Customer Loyalty:** Customers were segmented into New, Returning, and Loyal groups based on their total purchases.

	segments	segment_count
▶	Loyal	2339
	Returning	1137
	New	424

8. **Most Purchased Products by Category:** The most purchased products are Jewelry and Blouse, each with 171 orders.

	category	item_purchased	orders	ranking
▶	Accessories	Jewelry	171	1
	Accessories	Sunglasses	161	2
	Accessories	Belt	161	3
	Clothing	Blouse	171	1
	Clothing	Shirt	169	2
	Clothing	Dress	166	3
	Footwear	Shoes	150	1
	Footwear	Sandals	160	2
	Footwear	Boots	144	3
	Outerwear	Coat	161	1
	Outerwear	Jacket	163	2

9. **Subscription Status of Repeat Buyers:** A segment of repeat buyers (those with >5 previous purchases) subscribed, representing nearly one-third of all customers.

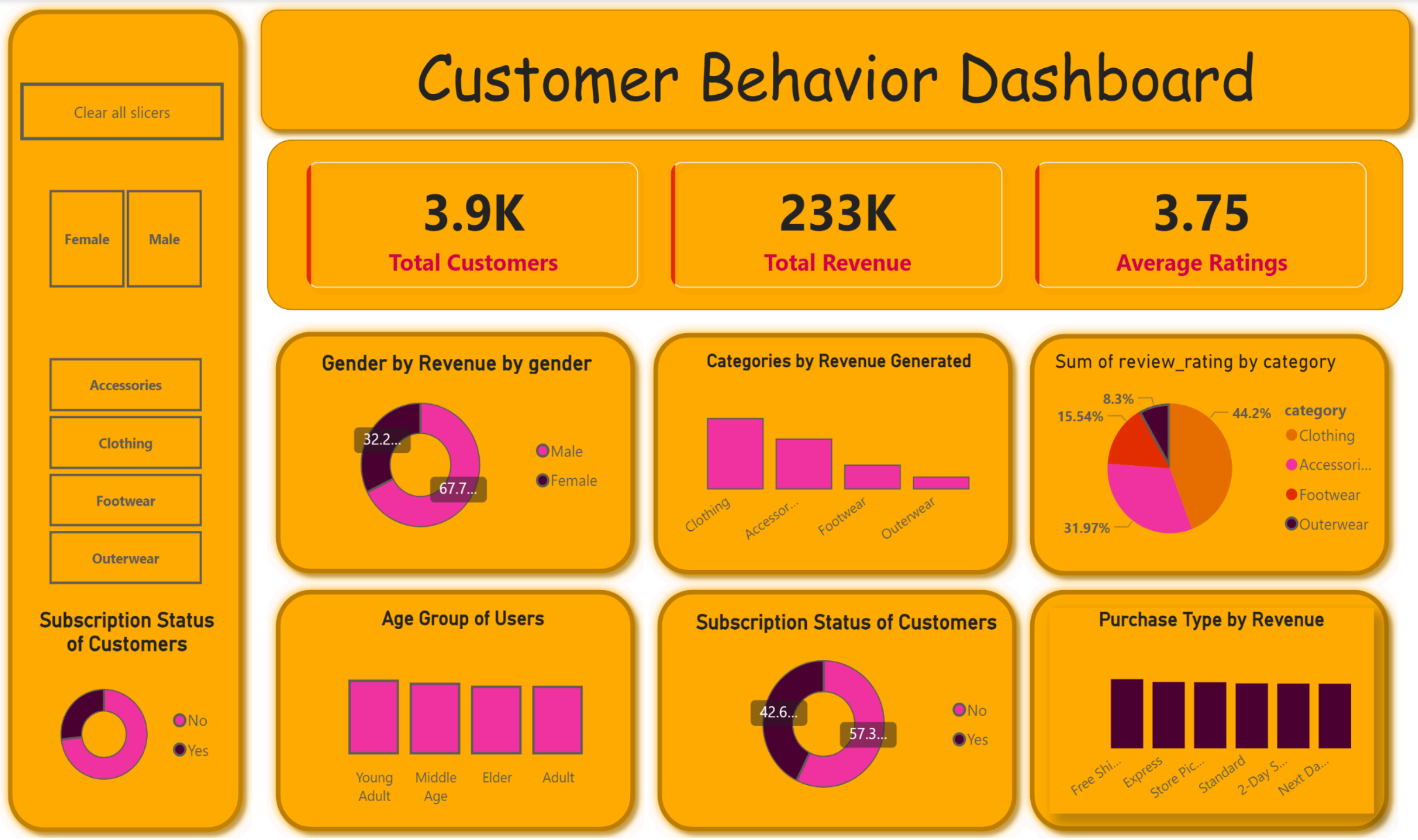
	class	customers
▶	Repeat Subscribers	958
	Others	2942

10. **Revenue Distribution of Age Group:** Young adults are the highest revenue generators among all age groups. However, the revenue difference is not substantial, indicating the importance of the other age groups as well.

	age_group	revenue
▶	Young Adult	62143
	Middle Age	59197
	Adult	55978
	Elder	55763

5. PowerBI Dashboard

For aesthetics and interactivity, we designed a PowerBI dashboard that gives a detailed summary of our dataset at a glance.



6. Recommendations

- **Subscription:** Most of the big spenders are non-subscribers. Targeted campaigns to convert these individuals are recommended.
- **Product Placement:** Clothing and accessories are the largest income-generating categories; more of these products should be placed on display.
- **Feedback Surveys:** Survey campaigns to get feedback from customers should be launched, as the overall average product rating of 3.75 on a scale of five (5) is acceptable but below the ideal target.
- **Advertisement:** Ad campaigns that target female shoppers with appropriate incentives should also be implemented.